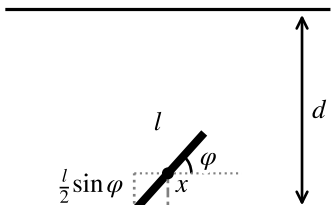
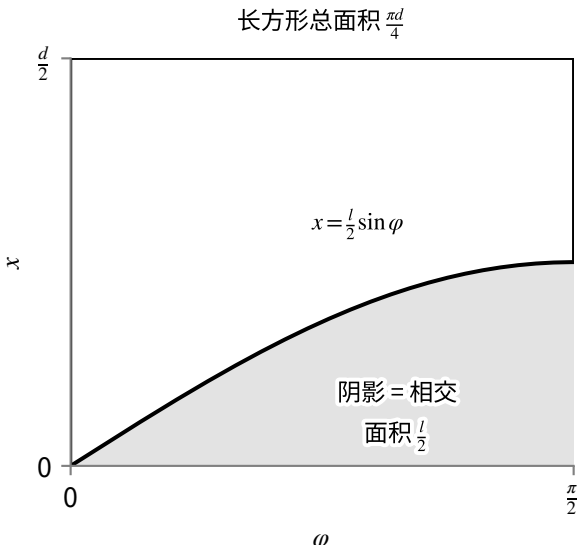
$x=\frac{l}{2}\sin\varphi$ 时恰好碰到